

# Circuit Simulation Lab:29

## Design of SISO

The purpose of this experiment is to design a 8 bit SISO. Serial In Serial Out (SISO) shift registers are a kind of shift registers where both data loading as well as data retrieval to/from the shift register occurs in serial-mode.

CODE:

```
module siso_8bit(
input clk,
input rst,
input serial_in,
output serial_out
);

reg [7:0] shift_reg;

always @(posedge clk)
begin

if(rst)
    shift_reg <= 8'b00000000;

else
    shift_reg <= {shift_reg[6:0], serial_in};

end

assign serial_out = shift_reg[7];

Endmodule

TESTBENCH
module tb_siso_8bit;

reg clk;
reg rst;
reg serial_in;

wire serial_out;

siso_8bit dut(
```

```

.clk(clk),
.rst(rst),
.serial_in(serial_in),
.serial_out(serial_out)
);

always #10 clk = ~clk;

initial begin

$dumpfile("siso.vcd");
$dumpvars(0,tb_siso_8bit);

clk = 0;
rst = 1;
serial_in = 0;

#20;
rst = 0;

// Send 10110011

serial_in = 1; #20;
serial_in = 0; #20;
serial_in = 1; #20;
serial_in = 1; #20;
serial_in = 0; #20;
serial_in = 0; #20;
serial_in = 1; #20;
serial_in = 1; #20;

// Extra clocks to observe output

serial_in = 0; #160;

$finish;

end

initial
begin

$monitor(
"time=%0t serial_in=%b serial_out=%b",

```

endmodule

